Supplementary Material

From Feedback to Revision: When Do Multi-Agent Roles Look Load-Bearing?

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Submission Id: 2850

What this document is

The main paper states every claim it makes and is complete without this file. What it cannot carry, at eight pages, is the full numeric backing behind some of its sentences. That backing is here.

Every table below is **generated** by a script from hash-manifested data — the frozen snapshot, or the separately manifested post-freeze runs (feedback-matched, forced submission) — except the single-agent comparator (Table 10), which comes from an earlier, separately collected corpus with no manifest here. No number in a table is hand-typed. Each table carries its own provenance header in the source, and the tables are generated by the same scripts from the same outputs as the main paper's, so a number cannot drift between the two.

Reading the denominators. Two populations recur and are not interchangeable. The *marginal* population is every trajectory in a cell, with each condition on its own denominator. The *paired* population is only those problems first-pass wrong under *both* conditions, on one shared denominator; that is the actual test of recovery. Recovery p values are exact McNemar on this population; first-pass and final-accuracy p values are exact McNemar on all problems, paired (as-graded contrasts on the problems graded under both); the pre-registered P4 contrasts use a bootstrap. Where a table shows both populations, they are labeled and must not be read across.

Recovery denominator. Throughout, recovery is conditioned on a recorded *first* evaluator verdict of incorrect — including trajectories that later died in regeneration, which score as failures under the pre-registered rule. The choice matters, and it is stated here so that a reader recomputing from the trajectories gets the same denominator.

1 The pre-registered endpoints in full

The main paper reports, in its Intervention section, that the pre-registered primary endpoint P4 — whether enforcement's effect on accuracy differs by actor — is null: none of the three pairwise difference-in-differences survives Holm; before correction the Llama-3.3–gemma-4 interval only just excludes zero, and a separate resample gives $p = 0.055$. It also reports that the P1a manipulation objective failed. Table 1 is the complete endpoint table promised there: all five endpoints as written before any intervention data existed, what each one returned, and the Holm-corrected verdict.

Two things in it deserve to be read rather than skimmed. First, only one of the three P4 contrasts is an *informative* null: the two involving Llama-3.3 cannot detect an interaction of any plausible size at their sample size, and are uninformative rather than evidence of absence. Second, the contrast this paper is organized around — role-structured versus role-free — was pre-registered as **secondary and exploratory**, and the feedback-matched role-free condition and the forced-submission run are outside the pre-registration (the feedback-matched condition's design and reading were committed in writing before its data existed). All are labeled exploratory in the main paper and here.

2 Functional collapse, all three measures

Table 2 is the full form of the main paper's Table 1: it adds the exact-byte-duplication lower bound, the structurally enforced condition, and a sensitivity analysis of the functional-collapse rate over three answer-marker definitions.

3 First pass and recovery, both populations

Table 3 gives the frozen comparison between the role-structured team and the role-free team with the feedback withheld, in both populations, so that the marginal rates and the paired test can be checked against each other. (The feedback-matched comparison is in the main paper's Table 2.)

For gemma-4 the accounting closes exactly. Both conditions reach a correct first answer on 1,284 of 1,600 problems, and a correct first answer is never lost (the loop halts on a passing verdict), so the paired final-accuracy difference, role-free minus role-structured — $-54/1{,}600 = -0.0338$, exact McNemar $p = 1.6 \times 10^{-5}$, conditioned on nothing — equals the difference in recovery counts, 16 minus 70. Stratified by the pair of first-pass outcomes, counting role-structured minus role-free final correct answers (so the strata sum to +54): the 1,205 problems both teams answer correctly first contribute 0; the 237 both miss contribute $38 - 4 = +34$ (discordant 37 versus 3); the 79 only the role-structured team misses contribute $32 - 79 = -47$; and the 79 only the role-free team misses contribute $79 - 12 = +67$. The last two strata condition on different problems, so their recovery rates (32/79 versus 12/79) are descriptive; together they contribute +20, in the same direction as the paired stratum. This accounting needs equal first-pass counts and does *not* transfer to the other two actors.

4 Cross-domain replication

Table 4 is the SciBench replication in full. The main paper carries both actors in Figure 2 and Table 2; this is the frozen comparison with its denominators.

The domain contract is *not* byte-identical to Omni-MATH, and the prompt files in the supplementary material make the difference checkable: of the 16 prompts the two domains share, 11 are byte-identical and 5 differ. The five are four of the six role-structured prompts (the Planner, Executor and Reviewer preambles and the Reviewer's routing rules) and the **evaluator**'s preamble, which



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Proc. of the 26th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2027), M. Baldoni, F. Fang, W. Yeoh, N. Yorke-Smith (eds.), 3–7 May 2027, Hanoi, Vietnam. © 2027 International Foundation for Autonomous Agents and Multiagent Systems (www.ifaamas.org).

Table 1: Every pre-registered endpoint, and what became of it. The five endpoints (pre-specified analyses) were fixed in writing before any intervention data existed. The intervention is structural enforcement (roles imposed by typed message formats) against the role-structured baseline (Planner, Executor and Reviewer prompts), per actor (the model playing all three agents); *fidelity* is whether the Planner abstains from computing (the P1a measure); *accuracy* is final accuracy over all problems with never-graded trajectories counted as incorrect. P3 tests, per actor, whether enforcement changes accuracy (exact McNemar test on the problems that exactly one condition gets right; the raw p, the p after Holm's step-down correction across the three actors, and the verdict at $\alpha = 0.05$). P4, the pre-registered primary, tests whether the enforcement effect differs between actors: each difference-in-differences is the effect in one actor minus the effect in another, with its 95% bootstrap confidence interval (resampling problems), the raw and Holm-corrected p, and $\text{MDE}_{80\%}$, the smallest interaction detectable with 80% power at the realized sample; a null is *informative* only where the MDE is below a plausible effect. P4 is null before and after Holm correction, and we report that null. P1 failed as written and was re-designated rather than dropped: the interface forbids the planner emitting an *answer*, and nothing in it stops the planner *computing*, a dissociation we report as a finding. P2 and P5 are reported as effects and associations and carry no p-value, so the testable contrasts are P3's three and P4's three. Holm is applied within each family of three; applied across all six, as the pre-registration's "Holm correction across P1–P5" would read, it changes no verdict. P4's intervals and p values come from two separately seeded bootstrap runs, so an interval can exclude zero while p is just above 0.05. The MDE is computed on problems shared by both actors, while each difference-in-differences uses each actor's own problem set, so neither can be derived from the other.

endpoint	disposition
P1 intervention → fidelity (also the manipulation check)	**failed as written; re-designated.** The interface forbids an *answer* field, not computation. Re-scoped to answer authorship, which it drives to 0.000 in all three actors (Table 9). Reported as a second dissociation, not as a pass.
P2 actor × interv. on fidelity	not estimable across the panel: the fidelity clauses are condition-specific, so no common scale exists. Reported as unresolved rather than substituted.

P3 intervention → accuracy (enforced − role-structured), Holm across three:

	raw p	Holm p	verdict
gemma-4-31B	0.0078	0.023	reject
gpt-oss-120b	0.60	0.85	retain
Llama-3.3-70B	0.43	0.85	retain

P4 actor × intervention on accuracy — *the pre-registered primary*, Holm across three:

	difference-in-diff. [CI]	raw p	Holm p	$\text{MDE}_{80\%}$	null is
gpt-oss-120b − gemma-4-31B	+0.0175 [-0.0050,+0.0406]	0.145	0.29	0.0325	informative
Llama-3.3-70B − gemma-4-31B	+0.0400 [+0.0006,+0.0800]	0.0553	0.166	0.0693	uninformative
Llama-3.3-70B − gpt-oss-120b	+0.0225 [-0.0181,+0.0631]	0.286	0.29	0.0748	uninformative

P5 Δfidelity vs Δoutcome	association only, pre-declared as non-causal: both sides are outcomes of the same intervention. No p-value is reported and none is corrected.

Only one of the three P4 nulls is informative. The largest per-actor main effect is small enough that a true interaction is plausibly ≤ 0.040; only the gpt-oss/gemma-4 contrast can see an effect that size. The two Llama-3.3 contrasts cannot detect an interaction of any plausible magnitude and are uninformative rather than evidence of absence.

must ask for scientific rather than mathematical equivalence. **Both role-free prompts are byte-identical across domains**, as are the six structurally enforced ones (enforcement was not run on SciBench).

That asymmetry is worth stating plainly rather than leaving to a diff: the role-structured prompts, whose team recovers less than the feedback-matched role-free team, are the ones that were rewritten, and the role-free prompts ran identical text in both domains. The rewrite is domain wording — "algebraic manipulation or numerical calculation" where Omni-MATH said "symbolic manipulation or arithmetic" — and the structural clauses are unchanged. A reader who suspects the adaptation is doing the work can check the files.

Topology, budget, decoding, retry policy and the evaluator protocol are unchanged. This is a replication of the *mechanism* under a matched protocol, not a rerun of identical prompts.

5 The feedback-matched control, run twice

The two gemma-4 cells of the feedback-matched role-free condition were run twice, independently, about thirteen hours apart, with byte-identical runner and prompt code; every other feedback-matched cell was run once. Each run also repeated the drift check of the main text on the same 400 frozen problems: the role-structured re-run reaches final accuracy 0.818 in the first and 0.845 in the second, against 0.825 frozen (paired $p = 0.75$ and $p = 0.20$). The main text reports the first run throughout. Table 5 gives both against the frozen role-structured cell. The ordering and the final-accuracy contrasts reproduce in both domains; the SciBench paired recovery test, on 100 problems, is $p = 0.013$ in the first run and $p = 0.052$ in the second.

Table 2: Functional collapse on the Omni-MATH benchmark, all three measures. The full form of the main paper's Table 1. Rows: the three actors (the model playing all three agents) under role-structured prompts (Planner, Executor and Reviewer), under role-free prompts (one generic prompt for all three agents) with the evaluator's feedback withheld, under the same role-free prompts with the feedback shown (run after the freeze), and under structural enforcement (roles imposed by typed message formats). *Functional collapse*: share of first cycles (one pass of the three agents) in which all three agents assert their own final answer. *Exact byte duplication*: share of first cycles in which two or more of the three messages are byte-identical, a lower bound on collapse (precision 1.000, recall 0.262 against the adjudication) that is never used to compare actors. *Adjudicated no division*: share of a judged sample of first cycles in which a model judge with a fixed rubric finds no division of labor; the judge did not see the grade but could infer the condition. Collapse is not applicable to the enforced condition, whose Planner schema has no answer field; for the role-free condition with feedback only functional collapse was measured.

actor	functional collapse	exact byte duplication	adjudicated no division
role-structured			
gemma-4-31B	0.000	0.000	0.000
gpt-oss-120b	0.120	0.000	0.000
Llama-3.3-70B	0.000	0.000	0.000
role-free, feedback withheld			
gemma-4-31B	**0.976**	0.000	**0.857**
gpt-oss-120b	**0.846**	0.680†	**1.000**
Llama-3.3-70B	**0.872**	0.008	**1.000**
role-free, with feedback			
gemma-4-31B	**0.974**	–	–
gpt-oss-120b	**0.883**	–	–
Llama-3.3-70B	**0.900**	–	–
structurally enforced			
all actors	n/a	≤ 0.001	0.071–0.188

Sensitivity. Under three answer-marker definitions the feedback-withheld role-free rate is 0.846–0.999; the role-structured rate is 0.000–0.238 under the two stricter markers and 0.328–0.892 under the loosest, which also counts any "therefore", "hence" or "= number" (as when a Planner restates givens), so read the role-structured level as "low," not a point estimate. The ordering holds in all 9 of 9 actor×marker cells.
†Not interpretable: 80% of this cell's planner messages are under 500 visible characters (the main paper's logging defect).

6 Where the role-structured Reviewer declines

The main paper's account of what roles do after feedback — a gate on submission — rests on Table 6, computed from the logged route of every Reviewer turn. A cycle ends in an approval (a submission) or a decline. Declines are split by when they happen: before the first submission, when no feedback exists; in a cycle that received a FAIL, the only cycle the role-structured clause covers; and in later cycles in which no clause forbids approval. Trajectories that never approve in six cycles are graded once at the end and never receive feedback; that grade is their first pass, so those graded wrong count as first-pass wrong and unrecovered.

On Omni-MATH the clause covers a minority of the declines of the role-structured Reviewer, and in the free cycles it approves far less often than the role-free team's third agent; gpt-oss is the closest of the three. On SciBench the clause covers more. With at most one round of feedback the role-free team recovers as much or more in every cell. Recoveries per verdict after the first are higher with roles in every cell, but that ratio compares different trajectories: most of the role-free team's later verdicts are further attempts on problems it has already failed more than once. The like-for-like comparison is the verdict-matched one in the main paper (Table 2): on the problems both teams first answer wrongly, with each role-free trajectory cut off after as many verdicts as the role-structured team received on the same problem, no cell shows a detectable difference. Section 7 tests how far that comparison can be pushed.

7 Robustness of the verdict-matched comparison

Matching on verdicts is coarse. When the role-structured team received a single verdict — it never submitted and was graded at the end, or it submitted once and stopped — the role-free trajectory cut to one verdict cannot recover either, and the pair ties at zero by construction. These forced ties are most of the gemma-4 and nearly all of the Llama-3.3 pairs (Table 7), so the pooled interval of the main paper is narrower than the information behind it. The cut is also one-sided: the role-structured count stops at that team's own passing verdict, which can only favor it. Table 7 therefore gives five variants: the problems with at least two role-structured verdicts; both trajectories cut at the smaller count; and the scoring defect recoded, each with and without Llama-3.3. No variant puts the role-free team significantly ahead: the two symmetric cuts favor it by 0.005 and 0.010, with intervals that include zero. On the informative problems the upper bounds reach +0.065 in favor of roles (+0.073 with the recoding), and with the recoding on all problems the role-structured team leads by +0.029 with an interval that excludes zero. The defensible reading is the one the main paper gives: matched on verdicts, the role-free advantage disappears, and any per-verdict advantage of roles is small. Verdict matching equalizes evaluator exposure, not calls: the role-structured team spends more cycles per verdict.

8 Removing the submission gate

The forced-submission run tests the gate directly. It uses the role-structured prompts and runner unchanged except for one step: after the Reviewer's turn, the pipeline submits the team's current answer to the evaluator whatever route the Reviewer chose, and the verdict and hint reach the agents through the same role-specific slots as in the role-structured condition. An offline test in the supplementary material (no model calls) checks that the prompts are byte-identical to the role-structured ones up to the first verdict and that the evaluator is called in every cycle. It was run once, after the freeze, for gemma-4 on the 400 Omni-MATH problems of the role-structured re-run, with the same endpoint, budget and evaluator. Table 8 compares it with that re-run, the frozen role-structured cell and the role-free team with feedback, restricted to the same

Table 3: First-pass accuracy and recovery, role-structured versus role-free (feedback withheld), on the Omni-MATH benchmark. Conditions: role-structured (Planner, Executor and Reviewer prompts) and role-free (one generic prompt for all three agents) with the evaluator's feedback withheld; per actor (the model playing all three agents). A trajectory is one problem's run through the pipeline; *first pass* is whether its first submitted answer is correct, and *recovery* is the share of first-pass-wrong trajectories (first verdict incorrect) that finish correct. The two panels have different denominators and must not be read across. *Upper:* first pass over the problems graded under both conditions (paired, as in the main paper); recovery over each condition's own first-pass-wrong trajectories (n in parentheses, computed per condition and therefore *not* a shared denominator); differences are role-free minus role-structured, with 95% bootstrap confidence intervals that resample problems. *Lower:* the paired population, only problems first-pass wrong under *both* conditions, one shared denominator n_{both}, which is the actual test; "role-struct. only" and "role-free only" count problems recovered under one condition and not the other, and the exact McNemar p tests whether those two counts differ. The recovery denominator counts every trajectory with a recorded incorrect first verdict, including those that later died in regeneration (a malformed output exhausted its regeneration budget); these score 0 under the pre-registered rule, used throughout.

First pass paired; recovery on each condition's own denominator

	first pass			recovery | first verdict wrong		
actor	role-struct.	role-free	difference	role-struct. (n)	role-free (n)	difference
gemma-4-31B	0.802	0.802	+0.0000 [−0.0156, +0.0150]	0.222 (316)	0.051 (316)	−0.1709 [−0.2247, −0.1171]
gpt-oss-120b	0.713	0.763	+0.0503 [+0.0339, +0.0666]	0.324 (463)	0.247 (380)	−0.0766 [−0.1361, −0.0153]
Llama-3.3-70B	0.308	0.316	+0.0077 [−0.0283, +0.0437]	0.014 (276)	0.015 (268)	+0.0004 [−0.0214, +0.0192]

Paired population — problems first-pass wrong under **both** conditions; one shared denominator

actor	n_{both}	role-struct.	role-free	role-struct. only	role-free only	exact p
gemma-4-31B	237	0.160	0.017	37	3	1.95×10^{-8}
gpt-oss-120b	327	0.242	0.202	52	39	0.208
Llama-3.3-70B	245	0.004	0.016	1	4	0.375

Recovery conditions on a post-treatment variable and is descriptive, never causal. No cell recovers before its first oracle verdict.

Table 4: The dissociation replicates out of domain: the SciBench benchmark. The complete 574-problem text-only slice of free-response college science, two actors (the model playing all three agents), 2296 trajectories, frozen separately from the Omni-MATH snapshot. Conditions: role-structured (Planner, Executor and Reviewer prompts) and role-free (one generic prompt for all three agents) with the evaluator's feedback withheld; the feedback-matched contrast is in the main paper's Table 2. *First pass*: whether a trajectory's (one problem's run) first submitted answer is correct, paired over all 574 problems; the exact McNemar p tests whether the problems right under only one condition outnumber those right under only the other. *Recovery*: the share of first-pass-wrong trajectories (first verdict incorrect) that finish correct, paired over only the problems first-pass wrong under *both* conditions (n_{both}), which is why its n is roughly a sixth the size; "r-s. only" and "r-f. only" count problems recovered under one condition and not the other. The two panels have different populations. Differences are role-free minus role-structured; pre-registered scoring (never-graded trajectories count as incorrect). gpt-oss's recovery cell cannot resolve its own effect at the realized n_{both}: its point estimate is in the same direction as Omni-MATH's, so it is underpowered rather than a non-replication. Llama-3.3 was not run in this domain.

	first pass (paired, all problems)					recovery (paired on first-pass wrong under both)						
actor	n	role-str.	role-free	diff.	exact p	n_{both}	role-str.	role-free	diff.	r-s. only	r-f. only	exact p
gemma-4-31B	574	0.794	0.794	+0.0000	1	99	0.212	0.020	−0.1919	21	2	6.60×10^{-5}
gpt-oss-120b	574	0.784	0.789	+0.0052	0.784	96	0.219	0.135	−0.0833	17	9	0.169

problems. Recovery rises to the role-free team's level while first-pass accuracy does not move; the Reviewer's logged approvals no longer decide anything.

9 Validation of the measures

The definition was frozen before the adjudication sheet was opened; 72 Planner messages were adjudicated by hand for P1a, each with a recorded reason. P1a agrees with the hand labels on 0.903 of messages (Cohen's $\kappa = 0.806$; precision 0.943, recall 0.868), clearing the pre-set 0.80 threshold. Agreement fell from 0.967 at $n = 30$ as adjudication continued; we report the final figure. Its residual errors point in opposite directions — symbolic results with no digits are missed, restated givens are flagged — so they do not push the measure consistently one way.

Table 5: The gemma-4 feedback-matched cells, run twice. Role-structured: Planner, Executor and Reviewer prompts (the frozen run). Role-free + feedback: one generic prompt for all three agents, with the evaluator's verdict and repair hint shown; both runs use byte-identical runner and prompt code, and the main text reports run 1. *First pass*: whether a trajectory's (one problem's run) first submitted answer is correct. *Recovery*: first-pass-wrong trajectories that finish correct, over each condition's own first-pass-wrong set (count / denominator). *Final*: accuracy at the end over all problems, never-graded trajectories counted as incorrect. The *paired* columns compare each role-free run with the role-structured run on problems first-pass wrong under both (recovery) or on all problems (final): problems recovered, or finally correct, only without roles versus only with them, with the exact McNemar p for the two counts.

	first pass	recovery	paired recovery	final	paired final
Omni-MATH					
role-structured (frozen)	0.802	70/316 = 0.222	–	0.846	–
role-free + feedback, run 1	0.802	193/317 = 0.609	89 vs 4, $p = 6.2 \times 10^{-22}$	0.921	137 vs 17, $p = 1.7 \times 10^{-24}$
role-free + feedback, run 2	0.811	177/302 = 0.586	95 vs 7, $p = 7.8 \times 10^{-21}$	0.922	139 vs 18, $p = 2.4 \times 10^{-24}$
SciBench					
role-structured (frozen)	0.794	30/118 = 0.254	–	0.847	–
role-free + feedback, run 1	0.796	49/117 = 0.419	21 vs 7, $p = 0.013$	0.880	31 vs 12, $p = 0.0054$
role-free + feedback, run 2	0.784	54/124 = 0.435	19 vs 8, $p = 0.052$	0.878	29 vs 11, $p = 0.0064$

Table 6: Where the role-structured Reviewer declines, and what each allocation buys. RS: role-structured (Planner, Executor and Reviewer prompts); RF: role-free (one generic prompt for all three agents) with the evaluator's feedback shown. Every column is over first-pass-wrong trajectories (one problem's run whose first submitted answer was graded wrong), per actor (the model playing all three agents) and benchmark. A cycle is one pass of the three agents; it ends in a submission to the evaluator when the third agent approves and in a decline otherwise. *RS declining cycles* per trajectory, split into those before the first submission (no feedback yet), those in a cycle that received a FAIL verdict (where the role-structured clause forbids approval) and the remaining "free" cycles; *clause share* is the FAIL column as a fraction of all three. *RS never submits*: trajectories with no approval in six cycles, graded once at the end. *Free-cycle approval*: the share of free cycles ending in approval. *By 2nd verdict*: the share recovered (finishing correct) with at most one round of feedback, over all first-pass-wrong trajectories. *Per later verdict*: recoveries per verdict after the first, a ratio over different trajectories in the two conditions (the verdict-matched comparison of the main paper's Table 2 is the like-for-like one). *Approvals*: submissions per trajectory.

	RS declining cycles			clause	RS never	free-cycle approval		by 2nd verdict		per later verdict		approvals	
actor, domain	before	FAIL	free	share	submits	RS	RF	RS	RF	RS	RF	RS	RF
gemma-4, Omni-MATH	2.24	0.81	1.20	0.19	84/316	0.25	0.96	0.206	0.438	0.41	0.28	1.27	3.20
gpt-oss, Omni-MATH	1.33	1.17	0.71	0.37	51/463	0.53	0.68	0.246	0.296	0.27	0.22	2.10	3.53
Llama-3.3, Omni-MATH	4.49	0.25	0.88	0.04	205/276	0.01	0.82	0.014	0.092	0.25	0.08	0.32	2.71
gemma-4, SciBench	0.55	1.38	1.48	0.40	3/118	0.35	0.99	0.237	0.239	0.26	0.16	1.95	3.62
gpt-oss, SciBench	0.41	1.53	0.60	0.60	1/124	0.67	0.78	0.258	0.317	0.19	0.16	2.68	3.95

10 An early campaign stopped for a truncation confound

The main run was gated on a smoke test requiring non-zero boundary rejections with reasons logged, answer-bearing calls equal across conditions, and a pinned backend. An early campaign was stopped after 121 trajectories when 74% of enforcement rejections proved to be the model cut off at the token limit mid-JSON rather than violating a role boundary. Truncation was not condition-specific but its *cost* was — a prose condition tolerates a truncated message where the enforced one rejects unparseable JSON — so left alone it would have produced a convincing "enforcement lowers accuracy" artifact. The limit was doubled and the pre-fix rows were never pooled.

11 Structural enforcement: both scorings, and the attrition that separates them

The main paper reports that enforcement loosens the submission gate in all three actors (and unlocks the answer for gemma-4 and Llama-3.3) but does not become end-to-end accuracy, and that the apparent accuracy benefit is survivorship. Table 9 is the evidence for both halves: the manipulation check, the one-sided attrition, and the same accuracy column under both scorings with intervals.

The enforced recovery and first-pass rates **do not compose** into the accuracy column. They cannot: trajectories that never receive a first verdict have no first pass, and those that die after one stay in the recovery denominator, so the three quantities do not share a denominator and the (1 − first-pass) × recovery identity does not hold. The pre-registration fixed in advance that unparseable trajectories count as incorrect rather than being dropped, and that

Table 7: Robustness of the verdict-matched recovery contrast. RS: role-structured (Planner, Executor and Reviewer prompts); RF: role-free (one generic prompt for all three agents) with the evaluator's feedback shown; a cell is one actor (the model playing all three agents) on one benchmark. The contrast takes the problems both teams first answer wrongly and cuts the RF trajectory (one problem's run) after as many evaluator verdicts (PASS or FAIL decisions) as the RS trajectory received on that problem, then compares recovery (finishing correct). Where RS received a single verdict (it never submitted again, or submitted once and stopped), the cut RF trajectory cannot recover either, so the pair ties at zero by construction. *Upper block*: such forced ties per cell, how many of them never submitted, the problems left, and the trajectories whose held answer matched the key but were graded wrong (the scoring defect of the main paper's Limitations). *Middle block*: the pooled difference (RS minus RF, stratified by cell, 95% bootstrap confidence interval resampling problems) under six definitions, with and without Llama-3.3; *equivalent* means the 90% interval lies inside ±0.05. Recoding counts an RS trajectory with a correct held answer as recovered, and an RF one only if it was not cut. *Lower block*: per-verdict pass rates, the share of trajectories whose k-th verdict is PASS among those whose earlier verdicts all failed, pooled over the five cells.

cell	paired n	forced ties	never submitted	left	held-answer recodes (RS / RF)
gemma-4, Omni-MATH	236	141	66	95	12 / 3
gpt-oss, Omni-MATH	319	84	36	235	4 / 0
Llama-3.3, Omni-MATH	250	242	187	8	2 / 1
gemma-4, SciBench	100	29	2	71	2 / 0
gpt-oss, SciBench	93	11	1	82	0 / 0

	all five cells			without Llama-3.3	
definition	n	difference [95% CI]	equivalent	n	difference [95% CI]
as in Table 2	998	+0.009 [−0.014, +0.033]	yes	748	+0.011 [−0.020, +0.043]
role-structured team received ≥ 2 verdicts	491	+0.018 [−0.029, +0.065]	no	483	+0.017 [−0.031, +0.064]
both cut at the smaller verdict count	998	−0.005 [−0.029, +0.019]	yes	748	−0.008 [−0.040, +0.024]
both cut, ≥ 2 verdicts	491	−0.010 [−0.059, +0.039]	no	483	−0.012 [−0.062, +0.037]
scoring defect recoded	998	+0.029 [+0.004, +0.054]	no	748	+0.035 [+0.003, +0.067]
recoded, ≥ 2 verdicts	491	+0.026 [−0.020, +0.073]	no	483	+0.025 [−0.023, +0.072]

per-verdict pass rate	role-structured	role-free with feedback
verdict 2	131/491 = 0.267	212/946 = 0.224
verdict 3	19/228 = 0.083	89/700 = 0.127
verdict 4	8/76 = 0.105	39/439 = 0.089

rule is what makes the as-graded reading survivorship rather than a finding.

The context-restored control. Restoring shared history at fixed enforcement (gemma-4 and gpt-oss, 400 problems each; attrition 4.5% and 9.8%) changes final accuracy relative to enforcement by +0.0450 [+0.0125, +0.0775] ($p = 0.0079$) for gemma-4 and by −0.0125 [−0.0475, +0.0200] ($p = 0.56$) for gpt-oss under the pre-registered scoring, and by +0.0196 [−0.0028, +0.0419] ($p = 0.14$) and −0.0089 [−0.0357, +0.0149] ($p = 0.65$) as graded (exact McNemar, paired).

12 The single-agent comparator

Table 10 is the single-agent reference under the same call cap. The main paper uses it as a guard against reading "collapse costs nothing" as "the system was one agent all along", and reports both directions.

The comparator comes from an **earlier, separately collected corpus whose two actor arms were produced in separate runs**. Each direction within an actor is therefore a sound guard, and *the gap between actors is not.* The result cuts both ways: a collapsed gpt-oss team beats its own single agent, while gemma-4's is significantly worse in both conditions. Neither "multi-agent systems are just single agents" nor "more agents help" survives contact with it.

13 Accumulated context as a competing explanation

Accumulated context is one candidate explanation of the role-structured team's recovery advantage over the feedback-withheld team; Table 11 stratifies by it.

It is reported with a caveat that is not decoration. Quartile cut points are computed over both conditions *pooled*, and accumulated context is nearly a function of condition — the role-free team accumulates less because it stops reworking. The strata are therefore not balanced, and this comparison cannot separate context from condition. It is **directional only**, and it is included because it does not behave uniformly: the gap survives in the lower quartiles, but in gpt-oss's top quartile the ordering reverses (as, on a handful of events, in Llama-3.3's third).

14 AI assistance

Reproduced in full here rather than pointed at. AAMAS requires that supplementary material ship inside the submission, and a disclosure a reviewer cannot open is not a disclosure.

AAMAS asks that where AI-assisted technologies contribute to hypotheses, methodology or experimental design, authors disclose the tool, version and role. This work used them substantially in all three, so what follows is detailed rather than pro forma.

Table 8: Removing the submission gate. gemma-4 (the model playing all three agents) on the Omni-MATH benchmark, the same 400 problems in every row. Role-structured: Planner, Executor and Reviewer prompts, in the frozen snapshot and in a re-run after the freeze; role-free with feedback: one generic prompt for all three agents, with the evaluator's feedback shown. The forced-submission run ("every cycle submitted") uses the role-structured prompts byte for byte, but the pipeline submits the team's answer to the evaluator in every cycle (one pass of the three agents), whatever the Reviewer decides; its approvals are logged but do not decide submission. A trajectory is one problem's run; it is first-pass wrong when its first submitted answer is graded wrong. *First pass*: share of trajectories whose first submitted answer is correct. *Recovery*: first-pass-wrong trajectories that finish correct, over each row's own first-pass-wrong set (count / denominator). *Final accuracy*: over all problems, never-graded trajectories counted as incorrect. *Verdicts* and *Reviewer approvals*: per first-pass-wrong trajectory, including the first. The lower block pairs two conditions on the problems both first answer wrongly (recovery) and on all problems (final accuracy): problems solved only by the first condition / only by the second, with the exact McNemar p for the two counts.

condition	first pass	recovery	final accuracy	verdicts	Reviewer approvals
role-structured, every cycle submitted	0.768	57/93 = 0.613	0.910	4.20	0.98
role-structured, re-run after the freeze	0.773	18/91 = 0.198	0.818	1.44	1.13
role-structured, frozen snapshot	0.795	12/82 = 0.146	0.825	1.41	1.15
role-free with feedback	0.795	45/82 = 0.549	0.908	3.34	3.34

		recovery (both first wrong)		final accuracy	
pair	n	first / second only	p	first / second only	p
every cycle submitted vs. re-run	72	26 / 2	3.0×10^{-6}	42 / 5	2.5×10^{-8}
every cycle submitted vs. role-free with feedback	70	7 / 7	1	8 / 7	1
re-run vs. role-free with feedback	65	3 / 22	1.6×10^{-4}	6 / 42	1.0×10^{-7}

Table 9: Structural enforcement: roles imposed by typed message formats. Enforced against role-structured (Planner, Executor and Reviewer prompts) and, where stated, role-free (one generic prompt for all three agents, feedback withheld), per actor (the model playing all three agents) on the Omni-MATH benchmark. A trajectory is one problem's run through the pipeline; it is first-pass wrong when its first submitted answer is graded wrong. *Planner authors answer*: the share of planner messages that state a final answer, under enforcement and (in parentheses) when roles are merely deleted; enforcement drives it to zero in all three actors, so the intervention engaged (this is the manipulation check; the main paper's R5 gives the first-cycle rate). *Locked answer*: the share of first-pass-wrong trajectories whose answer string never changed, enforced and (in parentheses) role-free / role-structured. *Recovery*: the share of first-pass-wrong trajectories that finish correct, enforced and (role-structured in parentheses), on the denominator of Table 3. *Attrition*: the share of enforced trajectories that never finish (a malformed output exhausted its regeneration budget). *Final accuracy, enforced − role-structured*: the difference in accuracy over all problems, with 95% bootstrap confidence intervals; the two accuracy columns are different populations, and the attrition column is why they disagree. The pre-registered rule scores never-graded trajectories incorrect; the as-graded column drops them and is a *sensitivity analysis only*: its gains are conditional on parsing successfully, and enforcement is the only condition where that removes a material fraction of the data. Enforcement restores recovery for Llama-3.3; for the other two actors the direction is consistent but not resolvable against that attrition.

	planner authors	locked answer	recovery		final accuracy, enforced − role-structured	
actor	answer (role-free) manipulation check	(r-f./r-s.) first-pass wrong	(role-str.) first-pass wrong	attrition	pre-registered (primary)	as-graded (sensitivity)
gemma-4-31B	0.000 (0.986)	0.119 (0.845/0.288)	0.314 (0.222)	8.0%	−0.0225 [-0.0387,-0.0063]	+0.0190 [+0.0054,+0.0333]
gpt-oss-120b	0.000 (0.916)	0.092 (0.208/0.052)	0.395 (0.324)	9.9%	−0.0050 [-0.0219,+0.0112]	+0.0361 [+0.0215,+0.0507]
Llama-3.3-70B	0.000 (0.958)	0.072 (0.806/0.750)	0.139 (0.014)	17.0%	+0.0175 [-0.0200,+0.0550]	+0.0453 [+0.0030,+0.0876]

Attrition elsewhere is negligible: 0.0–0.8% with roles, 0.0–2.5% without. The as-graded column turns every sign positive; the pre-registration fixed in advance that unparseable trajectories count incorrect, and **the benefit appears only after conditioning on successful parsing.**

Tools. A large language model agent (Anthropic Claude, Opus-class) performed research execution: experiment implementation, campaign operation, analysis scripting, manuscript drafting and internal review. Exact build identifiers were not retained for every session. A frontier model was additionally used as a *measurement instrument* for the rubric-based adjudication of whether a division of labor was realized; that model and rubric version are recorded and will be released with the data after review. The actor models under study are the experimental subjects, not assistance, and are named in the paper.

What the agents did. The hypotheses tested here — whether role organization contributes to first-pass generation and to post-failure revision, and through which channel — were formed with

Table 10: Against a single agent under the same call cap. Final accuracy over all problems (n) of each three-agent team, role-structured (Planner, Executor and Reviewer prompts) or role-free (one generic prompt for all three agents, feedback withheld), against a single agent of the same actor (the model playing all three agents) solving the same problems under the same cap on model calls; the difference is team minus single agent, with its 95% bootstrap confidence interval (resampling problems) and exact McNemar p. Both directions must be read together: a collapsed gpt-oss team beats its own single agent, so three agents emitting one answer is not functionally one agent; but gemma-4's team is worse than its own single agent in both conditions. Provenance: this panel is not from the frozen snapshot; the single-agent trajectories come from an earlier, separately collected trace corpus, joined by problem. Its two actor arms were produced in separate runs, so each direction is a sound directional guard while *the gap between actors is not*; no statement about relative magnitude across actors is supported here. *Realized call budget*: calls per problem to the evaluator and to the actor; the single agent makes about half the actor calls, and more evaluator calls than the role-structured team but fewer than the role-free one.

organization	team	single agent	difference
gemma-4-31B			
role-structured	0.8462	0.8688	−0.0225
	$[-0.0400, -0.0056]$, $p = 0.0122$, $n = 1600$		
role-free	0.8125	0.8688	−0.0563
	$[-0.0737, -0.0394]$, $p = 1.22 \times 10^{-10}$, $n = 1600$		
gpt-oss-120b			
role-structured	0.8031	0.7900	+0.0131
	$[-0.0044, +0.0300]$, $p = 0.152$, $n = 1600$		
role-free	0.8194	0.7900	+0.0294
	$[+0.0125, +0.0456]$, $p = 0.000678$, $n = 1600$		

Realized call budget

actor	protocol	evaluator	actor
gemma-4-31B	single agent	1.66	3.16
gemma-4-31B	role-structured team	1.11	6.76
gemma-4-31B	role-free team	1.95	5.87
gpt-oss-120b	single agent	1.69	3.40
gpt-oss-120b	role-structured team	1.35	7.54
gpt-oss-120b	role-free team	1.85	6.25

Table 11: Recovery by pooled accumulated-context quartile, role-structured versus role-free (feedback withheld). Conditions: role-structured (Planner, Executor and Reviewer prompts) and role-free (one generic prompt for all three agents) with the evaluator's feedback withheld; per actor (the model playing all three agents) on the Omni-MATH benchmark. *Recovery*: the share of first-pass-wrong trajectories (one problem's run whose first submitted answer was graded wrong) that finish correct, with n the number of such trajectories in the quartile; the difference is role-structured minus role-free. *Context* is the total character count of all Planner, Executor and Reviewer messages in the trajectory; quartile cut points are computed *once per actor over both conditions pooled*. This is directional only. Accumulated context is nearly a function of condition (the role-free team talks less because it stops reworking), so the strata are not balanced and this comparison cannot separate context from condition. It is reported because accumulated context is a candidate explanation of the role-structured team's advantage over the feedback-withheld team, and because it does not behave uniformly: the gap survives in the lower quartiles, but in gpt-oss's top quartile the ordering reverses (as, on a handful of events, in Llama-3.3's third).

	role-structured		role-free		
context quartile	recovery	n	recovery	n	difference
gemma-4-31B					
Q1 (least context)	0.662	65	0.151	93	+0.511
Q2	0.280	50	0.009	108	+0.271
Q3	0.047	85	0.014	73	+0.033
Q4 (most)	0.078	116	0.000	42	+0.078
gpt-oss-120b					
Q1 (least context)	0.923	13	0.258	198	+0.665
Q2	0.656	128	0.293	82	+0.363
Q3	0.262	164	0.191	47	+0.071
Q4 (most)	0.070	158	0.189	53	−0.119
Llama-3.3-70B					
Q1 (least context)	0.100	40	0.031	96	+0.069
Q2	0.000	86	0.000	50	+0.000
Q3	0.000	103	0.030	33	−0.030
Q4 (most)	0.000	47	0.000	89	+0.000

the agents' assistance and then subjected to pre-registered and exploratory tests. The pre-registered primary hypothesis (P4) was not supported, and the paper reports that null. Condition design, budget matching, the pre-registration and the stopping and exclusion rules were drafted by agents under human direction and frozen before confirmatory analysis. Agents wrote the runner, operated and froze the campaign, wrote all analysis code, drafted and revised the manuscript, and conducted several independent internal reviews — including blind reviews given only the compiled PDF, and adversarial audits tasked with invalidating the paper's own claims. Every table is generated by script from hash-manifested data (the frozen snapshot or the separately manifested post-freeze runs; the single-agent comparator excepted), and every number in the prose was checked against the scripts' outputs.

What the human authors directed and approved. Research direction and every pivot; the decision to pre-register and the predeclared not-submit conditions; the choice of actor models and domains; the decision to use model adjudication in place of human annotation, with the instruction not to overstate what it establishes; the instruction that the page limit may not be met by hiding caveats; and final approval of all claims. The scope limits — not overstating actor generality, not publishing the data during review — were human decisions.

Prompts, and what was not retained. The exact role prompts, schemas and evaluator prompt are included in the supplementary material. Session prompts directing the agents' research work were **not retained verbatim**, and we do not reconstruct prompts that were not kept; standing instructions are reconstructed in the project record from the decision log. Model build identifiers were not retained for every session. Everything load-bearing for a reported number — experimental prompts, schemas, configurations, budgets, and the frozen rows themselves — is retained; the code and manifests are in the supplementary material, and the trajectories will be released after review.

Defects disclosed in the paper. Agent review of the instrumentation and analysis surfaced three properties of the data that the paper reports rather than repairs: a logging defect that left one actor's planner output unrecorded; a scoring defect that inflates the main contrast, reported with a recoded sensitivity analysis; and a renamed role token visible to the adjudicator, which is why the paper describes its adjudication as outcome-blind but not condition-blind.